MODULE 06 · SECTION 1 - FUNDAMENTALS OF GEN AI

The ML lifecycle on Google Cloud

Five stages from raw data to a model in production, and the Google Cloud tool for each.

SECTION 1 of 4 EXAM WEIGHT ~30% MODULE 6 of 33

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The model is the smallest part of an ML system. **The lifecycle around it** decides whether it succeeds.

The ML lifecycle in five stages

01
Ingest
Bring data in from where it is created.

02
Prepare
Clean, join, label and shape it.

03
Train
Teach a model, or adapt an existing one.

04
Deploy
Put the model into service.

05
Manage
Version, monitor and improve it over time.

ON THE EXAM

These five stage names come straight from the exam guide. The questions map a tool to a stage.

Ingest: getting data in

TOOL	BRINGS IN
Pub/Sub	Streams of events as they happen: clicks, sensor readings, stock updates
Cloud Storage	Files of any kind: documents, images, audio, video, exports
BigQuery	Large volumes of structured data for analysis, loaded or streamed
Cloud SQL and other databases	Live operational records, such as orders and accounts

THE TIEBREAKER

Streaming and in real time point to Pub/Sub. Files point to Cloud Storage. Analysis at scale points to BigQuery.

Prepare: making data fit to learn from

TOOL	DOES
BigQuery	Cleans, joins and reshapes data with SQL, at any scale
Dataflow	Transforms batch and streaming data in automated pipelines
Data Labeling	Organises human labelling of images, text and video
Feature Store	Keeps prepared features consistent for training and serving

THE TAKEAWAY

Preparation usually takes more time than training. Plan the budget and the people accordingly.

Train: teaching or adapting a model

TOOL	CHOOSE IT WHEN
AutoML	You have labelled images or tables, and want a model without writing code
BigQuery ML	Analysts want to build models using SQL, where the data already lives
Managed Training	Data scientists need full control with their own code
Model Garden, then tuning	A foundation model nearly fits; adapt it rather than build

THE TIEBREAKER

No code and labelled data points to AutoML. Full control points to custom training.

Online or batch inference?

ONLINE INFERENCE

One answer, right now

- A live endpoint answers each request
- Someone or something is waiting
- Checkout, chat, fraud checks
- Always on, so always costing

BATCH INFERENCE

Many answers, later

- A large job scores everything at once
- Nobody is waiting for each result
- Monthly scores, overnight reports
- Runs only when needed, cheaper

THE TIEBREAKER

Is anyone waiting for the answer? If yes, online. If no, batch.

Manage: keeping models trustworthy

TOOL	DOES
Model Registry	One place to store, version and track your models, and deploy them
Pipelines	Automates the lifecycle so it runs the same way every time
Model Monitoring	Watches live models for drift and skew, and raises alerts
Evals	Scores generative AI output for quality, against rubrics

ON THE EXAM

Which version is live, and who changed it? That is a Model Registry question.

Agent Platform: one home for the lifecycle

Model Registry
Versions
Metadata

Model Garden
Discover models
Google and partners

Pipelines
Repeatable
workflows

AutoML
No-code training
Images and tables

Agent Platform
Gemini Enterprise Agent Platform
Formerly Vertex AI
One unified platform for ML and gen AI

Managed Training
Your own code
Full control

Model Monitoring
Drift and skew
Alerts

Endpoints
Online and batch
inference

Feature Store
Consistent
features

ON THE EXAM

When a question wants one platform to build, deploy and manage models, the answer is Agent Platform, or Vertex AI on older papers.

Model Garden or Model Registry?

MODEL GARDEN

Models others built

- A catalogue of 200+ models
- Google, partner and open models
- Discover, test, customise, deploy
- Where you start

MODEL REGISTRY

Models you own

- Your trained and tuned models
- Versions, metadata, lineage
- Track and deploy them
- Where you keep order

THE TIEBREAKER

Finding a model to use is Garden. Keeping your own versions in order is Registry.

Why the manage stage never ends

- **The world changes:** customers, prices, suppliers and language all shift.
- **Drift:** live data stops looking like the training data, and accuracy slips quietly.
- **Retraining** on fresh data brings the model back in line.
- **MLOps** applies DevOps habits to ML: automate, version, monitor, repeat.
- **On Google Cloud:** Pipelines, Model Registry and Model Monitoring together.

> THE TAKEAWAY
>
> A model is never finished. Budget for running and retraining it, not just for building it.

The lifecycle when you use a foundation model

- **Ingest and prepare** still matter: the documents you ground answers in.
- **Train** usually shrinks to choose a model, then prompt, ground or tune.
- **Deploy** may mean an app, an agent, or an API behind your product.
- **Manage** adds new work: evaluating answers and watching for model upgrades.
- **Skipping management** is how a good pilot becomes a bad product.

> THE TIEBREAKER
>
> Gen AI shortens the training stage. It does not remove the other four.

Wrenfield's delivery-days model, stage by stage

STAGE	WHAT WRENFIELD DOES
Ingest	Orders from Cloud SQL and courier updates via Pub/Sub, into BigQuery
Prepare	Cleans and joins 200,000 past deliveries in BigQuery
Train	AutoML on the tabular data, with no code
Deploy	An online endpoint, so checkout shows a delivery estimate
Manage	Registry holds each version; Monitoring watches for drift

THE TAKEAWAY

One model, five stages, six Google Cloud tools. The model training is the shortest step.

> An insurer's data science team has built dozens of model versions over two years. They sit in different storage folders, with notes in spreadsheets.
>
> Nobody can say for certain which version is live. The team needs one place to version models, track their metadata, and deploy them.

Which Google Cloud capability should they use?

A Better-organised Cloud Storage folders

B Agent Platform Pipelines

C Model Garden

D Model Registry on Agent Platform

◆ PAUSE AND DECIDE

Which Google Cloud capability should they use?

A Better-organised Cloud Storage folders

Stores files, but no versions, metadata tracking or deployment.

B Agent Platform Pipelines

Automates workflows. It is not the catalogue of versions.

C Model Garden

A catalogue of other people's models, not yours.

D Model Registry on Agent Platform

Versions, metadata and deployment for your own models.

THE SIGNAL

Version, track metadata, deploy, one place: the wording of Model Registry.

> Wrenfield wants stock levels from its 118 stores fed to a replenishment model as sales happen, not overnight.
>
> Each till and stock scanner sends a small update every time an item sells or arrives.

Which service should ingest these updates?

A Pub/Sub

B A nightly export to Cloud Storage

C A weekly scheduled load into BigQuery

D Data Labeling

◆ PAUSE AND DECIDE

Which service should ingest these updates?

A Pub/Sub

Ingests streams of small events as they happen.

B A nightly export to Cloud Storage

Overnight is exactly what they want to avoid.

C A weekly scheduled load into BigQuery

Even slower. BigQuery may store the data later.

D Data Labeling

Organises human labels. It ingests nothing.

THE SIGNAL

As sales happen, small updates from many sources: streaming ingestion, so Pub/Sub.

> Once a month, Wrenfield Pay scores all 410,000 cardholders for the chance of falling behind, to plan a support mailing.
>
> Nobody waits for any individual score. The results are needed by the next morning.

How should the model be used?

A Batch inference across all cardholders

B An always-on online endpoint called once per cardholder

C Retrain the model from scratch each month instead

D Model Monitoring to produce the scores

◆ PAUSE AND DECIDE

How should the model be used?

A Batch inference across all cardholders

One large job, nobody waiting, ready by morning.

B An always-on online endpoint called once per cardholder

Works, but pays to stay on all month for one job.

C Retrain the model from scratch each month instead

Training is not scoring. It produces no predictions.

D Model Monitoring to produce the scores

Monitoring watches a model. It does not make predictions.

THE SIGNAL

Nobody waiting, everyone scored at once: batch.

> Wrenfield's delivery-days model was accurate at launch. Three months ago, Wrenfield changed its main courier.
>
> Estimates have been getting steadily worse, and nobody noticed until complaints rose. The CFO asks what should have been in place.

What was missing from the setup?

A. A larger, more capable model at launch

B. More input columns in the original training data

C. A newer model from Model Garden

D. Model Monitoring with drift alerts, and a retraining pipeline

◆ PAUSE AND DECIDE

What was missing from the setup?

A. A larger, more capable model at launch
It was accurate at launch. The world changed afterwards.

B. More input columns in the original training data
Old data cannot describe a courier change that came later.

C. A newer model from Model Garden
Garden models know nothing about Wrenfield's couriers.

D. Model Monitoring with drift alerts, and a retraining pipeline
Spots the drift early, then refreshes the model on new data.

THE SIGNAL

Accurate at launch, worse after a change, unnoticed: drift without monitoring.

Module 6 in five lines

- Five stages: ingest, prepare, train, deploy, manage. It is a loop, not a line.
- Ingest: Pub/Sub for streams, Cloud Storage for files, BigQuery and databases for records.
- Train: AutoML with no code, BigQuery ML with SQL, Managed Training for full control.
- Deploy online when someone waits; batch when nobody does.
- Manage: Model Registry for versions, Monitoring for drift, Pipelines to repeat.